assigment Q ans

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1 Question 4.5

Total mass of plate+cake = 0.5 + 1.8=2.3kg

$$N = mg = 2.3 \times 9.8 \approx 22.54 \textbf{ N}$$

$$f_{smax} = 0.4 \times 22.54 = 9.02 \textbf{ N}$$

since applied force = 12 N

$$12 > 9.02$$

So clearly the plate will start moving and since plate started motion we will deal with kinetic friction.

$$f_k = 0.3 \times 22.54 \approx 7.76 \textbf{ N}$$

$$F_{net} = 12 - 7.76 \approx 5.54 \textbf{ N}$$

$$a = \frac{F}{m} = \frac{5.54}{2.3} = 2.28 m/s^2$$

Using SUVAT

$$S = u + \frac{1}{2}at^2$$

starting from rest so u=0

$$S = \frac{1}{2} \times 2.28 \times 3^2 = 10.26 \textbf{ m}$$